

Biodiversity

Hot spots: Significance, mega-biodiversity

The term “biodiversity hotspot” was first introduced by British Biologist **Norman Myers** in 1988. A biodiversity hotspot is a biogeographic region with significant levels of biodiversity that is threatened by human habitation.

Criteria for recognizing **biodiversity hot spots**:

1. A region must have **at least 1500 vascular plants** as endemics.
2. It must have **>70%** of its original natural vegetation threatened.

An endemic species is a species that's found in a certain area and nowhere else on earth. To identify hotspot why plants are so important? We know Plants are the primary producers. Animals go where the plants are. Plants are the base of food webs. Life attracts other life and it depends on other life. That's why the plants are very important. At the moment Conservation International formally recognizes 36 biodiversity hotspot areas on earth. The interesting thing about this is that less than three percent of the earth's land surface area is represented by these hotspots.

What is the Significance of hotspots?

- Biodiversity is the building blocks of all life on earth. Without species, there would be no air to breathe, no food to eat, no water to drink. There would be no human society at all. And as the places on Earth where the most biodiversity is under the most threat, hotspots are critical to human survival.
- There would be no life on Earth without biodiversity, making these biodiverse hotspots, even more critical for our survival
- The maps of hotspots overlap with maps of natural places that most benefit people.
- That's because hotspots are among the richest and most important ecosystems in the world — and they are home to many vulnerable populations who are directly dependent on nature to survive. By one estimate, in spite of containing 2.5% of Earth's land surface, the forests, wetlands and other ecosystems in hotspots account for 35% of the “ecosystem services” that vulnerable human populations depend on.

What are the most threatened hotspots?

- Most hotspots are located in tropical forests.
- **Atlantic forest, Brazil:** 20,000 plant species, about half of which, are endemic. Of the original 1.2 million km², only 8% remains.
- **Polynesia-Micronesia, South Pacific Ocean:** It includes coastal wetlands, coral atolls, savannas and tropical rainforests.



Fig: - Atlantic forest, Brazil

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- Some other notable hotspots are **Columbia**, which has the highest rate of species by area unit worldwide and the largest number of endemics; ~20% species can be found here.

Mega-biodiversity:

India is one among the seventeen ‘megadiversity’ countries in the world, a concept which was introduced by R.A. Mittermier and T.B. Vernier. Megadiversity is a much less discussed subject than biodiversity. This term and another term ‘Hot Spots’ have recently been used by World Bank and other World bodies for species diversity and endemism in the World’s selected few rich floral and faunal zones. “Just as the G-7 countries concentrate a major portion of the world’s economic wealth, the 17 Megadiversity Countries have within their borders more than two thirds of our planet’s biological wealth, its biodiversity,” explains Conservation International’s President Dr. Russell A. Mittermeier.

The Megadiversity concept was created in an attempt to prioritise conservation efforts around the world. More than half of the world’s forests have already disappeared, and more are destroyed each year. Megadiversity is not only a concept, it is a call for action to ensure the survival of all forms of life on earth. Two spots identified as ‘Megadiversity’ and ‘Hot Spots’ in India are North-eastern

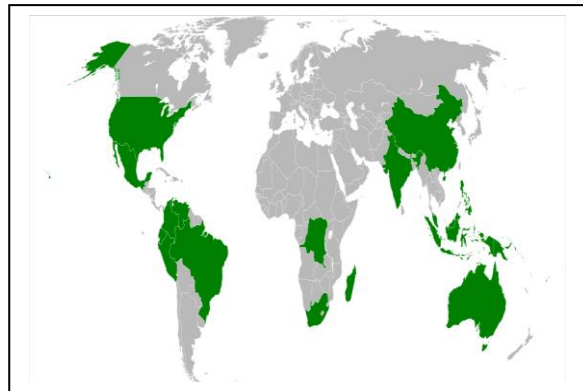


Fig: - Megadiversity of world

Himalayas and Western Ghat. But India as a whole has been marked a megadiversity area. Indians are not yet very much conscious and concerned about biodiversity loss and degradation of entire ecosystem. As the conservation need is urgent in the face of depletion India needs a well-designed strategy to protect these resources. The distribution of biodiversity in India is also important. India, which occupies just two percent (2.4%) of the total landmass of the world, harbors a rich biodiversity comprising about 8% of the known biodiversity of the world.

Conservation International identified **17 megadiverse countries** in 1998. Many of them are located in, or partially in, tropical or subtropical regions. Megadiversity means exhibiting great biodiversity. The main criterion for megadiverse countries is endemism at the level of species, genera and families. A megadiverse country must have at least 5,000 species of endemic plants and must border marine ecosystems. Nations that harbor most of Earth’s species and high numbers of endemic species.

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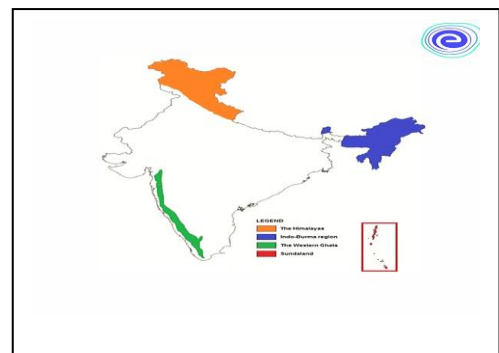
Biodiversity in India:

- **What makes India a mega-biodiversity nation?**
 1. Species richness
 2. Species endemism
 3. Biogeographically different regions
 4. Biodiversity Hot spots
 5. Biodiversity conservation efforts
- India has **diverse geographical features**, ranging from desert, mountains, highlands, tropical and temperate forests, swamp lands, plains and grasslands each spanning different climates.
- India has **23.39% of geographical area** under forest cover.
- 7.6% of mammalian, 12.6% of all avian, 6.2% of all reptilian, 4.4% of all amphibian, 11.7% of all fish and 6% of all flowering plant species.
- Within **2.4% of land area**, it accounts for nearly **7%** of recorded species and almost **18%** of the human population
- In terms of **species richness**, India ranks 7th, 9th in birds, 5th in reptiles.
- In terms of **endemism**, India is 10th in birds with 69 species, 5th in reptiles with 156 species and 7th in amphibians with 110 species

Biodiversity hotspots in India:

Overall, 36 biodiversity hotspot areas are there on earth. India contains **4 of the 36 biodiversity hotspots**, they are found in:

1. Western Ghats
2. Himalayas
3. Sundaland
4. Indo-Burma region



1. Western Ghats and Sri Lanka:

- The mountainous zones and the monsoons make a substantial contribution to the biodiversity of the Western Ghats which consists of a rich variety of plant, reptile, and amphibian species.
- It is among the **top 8 biodiversity hotspots** in the world.
- It has original reserve of 2 lakh km², only ~143,611 km² remain.
- 1600 km long chain of hills running along the western peninsular coast of India, including Kerala, Tamil Nadu and Karnataka.

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- Western Ghats is characterized by **heavy rainfall**, and contain moist deciduous and rain forests.
- The region is home to 450 birds (species), 140 mammals, 260 reptiles and 175 amphibians, now rapidly heading towards extinction.
- **How western ghats are Importance to us:** Any reduction in rainfall due to deforestation of the Western Ghats would lead to a warming of the peninsula as well.

2. Himalayas:

- The Himalaya Hotspot has some of the **highest peaks** in the world including Mt. Everest and K2. The varied topography of this hotspot supports a wide range of ecosystems like alluvial grasslands, subtropical broadleaf forests and alpine meadows.
- Significant feature of this hotspot is the occurrence of vascular plants at altitudes as high as 6,000 meters. Vultures, tigers, elephants, rhinos and wild water buffalo are some of the species found in this hotspot.
- Region comprising **Bhutan, Northeast India, southern, central and eastern Nepal.**
- It is geologically young and shows **high altitudinal variation.**
- Of the estimated 10,000 species of plants in the Himalayan hotspot, 3160 are endemic, as well as 71 genera.
- Despite icy zones starting at ~5500-6000 m, there are some species of vascular plants occurring at such high altitudes.
- Nearly 980 birds (15 endemic), 300 mammals (12 endemic), 175 reptiles (50 endemic), 105 amphibians (40 endemic) have been observed here.

3. Sundaland:

- Sundaland is one of the biologically richest hotspots on Earth, The Sundaland hotspot has over 25,000 plants, 2,000 species of orchids and some of the world's largest flowers belonging to the Rafflesia family.
- Sundaland hotspot covers the western half of the Indo-Malayan archipelago. It includes islands of Malaysia, parts of Thailand, Singapore. It is represented by the Andaman & Nicobar Islands from India.
- Of the original reserve of 1.5 million km², only ~1 lakh km² remain.
- They have a rich terrestrial and marine ecosystem comprising mangroves, coral reefs and sea grass beds, with a wide variety of flora and fauna.

Andaman and Nicobar Islands:

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- These constitute a group of **572 islands**, falling under the **Indo-Malayan biogeographic realm**, with Andaman resembling Myanmar and Thailand, while Nicobar was similar to Indonesia and South east Asia, and closer to the sundaland region.
- The islands harbour around **9130 animal species**, in terrestrial habitats, of which, 5859 are marine species.
- A high percentage of endemism (24.95%, 816 species) has been observed in terrestrial fauna, which is 4 times higher than marine habitat endemism.
- These **high rates of endemism** can be attributed to isolation of land masses, while the low rates of marine endemism can be due to continuity in the water medium.
- Some of the endangered species include **whales, dolphins, dugong, saltwater crocodile, hornbills, marine turtles, seashells of the Trochus species.**

4. Indo-Burma:

Indo-Burma: encompasses several countries, spread out from Eastern Bangladesh to Malaysia and includes Northeast India south of Brahmaputra River, covering ~ 2.4 million km² of tropical Asia, east of the Ganges-Brahmaputra lowlands.

- They include tropical and subtropical moist, dry and broadleaf forests, temperate and coniferous forests, mangroves, swamps and seasonally inundated grasslands.
- Most of this region is characterized by distinct seasonal weather patterns, such as cool, dry, northern winter months, and rains during spring as a result of Southwest monsoons.

